Curriculum Vitae/Resume

Yan Zhuang

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Education

Renmin University of China, B.Econ. in Economics of Resources and Environment — Sep. 2022 - Jun. 2026 (E)

- GPA: 3.66/4.0
- **Related Courseworks:** Econometrics, Intermediate Microeconomics, Intermediate Macroeconomics, Non-market Valuation

University of California, Davis, Global Study Program (Exchange) — Jan. 2025- Mar. 2025

- GPA: 4.0/4.0
- **Related Courseworks:** Environmental Economics, Intermediate Micro Theory: Imperfect Competition & Market Failure

Research Interest

- Environmental Economics, Energy Economics, and Climate Change
- Applied Econometrics and Causal Inference

Research Experience

Integrating Climate Change Impacts into the Assessment of China's Onshore Wind Power Potential — Jun. 2024 – Present

Co-authored with Dr. Yanhua Wang
Under Review at Environmental Impact Assessment Review

- Developed a multi-tiered assessment framework (Geographical-Technical-Economic) to project wind power potential up to 2100 using CMIP6 data (BCC-CSM2-MR/CAMS-CSM1-0/FGOALS-f3-L) under three SSP scenarios.
- Innovated the Levelized Cost of Energy (LCOE) model by incorporating the impact of temperature on turbine efficiency and precipitation-induced blade erosion on operational lifetime.
- Revealed a resource-demand mismatch and quantified regional economic risks, finding that precipitation degradation increases LCOE by >10% in southern coastal provinces, offsetting wind speed gains.

Sand Out, Water In: The (Un)intended Consequence of Afforestation on Hydrological Cycle and Agricultural Outcomes (Thesis) — Oct. 2024 – Present

School of Ecology and Environment, Renmin University of China

- Develop a quasi-experimental identification strategy exploiting high-frequency variation in prevailing wind direction to isolate spatial atmospheric spillovers of afforestation on precipitation and agricultural productivity.
- Construct novel wind-mediated forest exposure measures by combining satellite-based forest cover, Lagrangian air-mass trajectories, and reanalysis wind data to separately identify moisture advection and shelterbelt (wind-blocking) mechanisms.
- Provide causal evidence that upwind afforestation increases downwind wheat yields even conditional on local precipitation and weather controls, consistent with non-precipitation atmospheric moisture

channels.

- Quantify the aggregate economic value of afforestation-induced atmospheric externalities through a spatially explicit counterfactual analysis of China's Three-North Shelterbelt Program.

ABM-based Policy System Research on Express Packaging Waste Management Mar. 2024 – Mar. 2025

Project Leader
Awarded "Excellent Completion" distinction at Renmin University of China

- Led a team to analyze primary data from 1,600 respondents (consumers, merchants, couriers) to parameterize the utility functions and interaction rules of heterogeneous agents.
- Constructed an Agent-Based Model (ABM) to simulate dynamic behavioral responses to "Reduce, Replace, Recycle" policies, overcoming the limitations of static equilibrium models.
- Identified incentive compatibility mechanisms, finding that non-monetary incentives significantly outperform traditional subsidies for merchants in promoting packaging reduction.

2024 Tsinghua International SDGs Case Analysis Competition Apr. 2024 – Aug. 2024

Honorable Mention, Group Leader

- Analyzed stakeholder relationships under the Carbon Inclusive Mechanism framework, the factors influencing their actions, incentive mechanisms, and practical effects.
- Proposed Chinese solutions for SDG 12: Responsible Consumption and Production.

Honors, Scholarships & Assistantships

Teaching Assistantship, Carbon Market Theory and Practice Fall 2025

- Selected by Dr. Wang to support undergraduate coursework based on academic excellence.

Academic Perfection during Global Study Program Winter 2025

Academic Excellence Scholarship, Renmin University of China 2023, 2025

Professional Experience

Beijing Greenovation Institute for Public Welfare Development Aug. 2024 - Nov. 2024

Intern

- Monitored and summarized key proceedings from the United Nations Convention on Biological Diversity's 16th Conference of the Parties (COP16) for internal briefings.
- Conducted comparative research on the establishment and cooperation of national parks in China and the U.S..
- Contributed research on China-ASEAN agricultural and food systems cooperation to a published policy report.
- Authored a research brief on the inclusion of gender perspectives in nations' Intended Nationally Determined Contributions (NDCs).

Skills

Computing: R, Stata, Python, LaTeX

Languages: Mandarin Chinese (native), English (proficient)